

# Flood Detection from Aerial Imagery.

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Binary classification of post-disaster drone imagery using transfer learning with ResNet50 and ResNet101 backbones.

## DATASET

FloodNet-Supervised v1.0 — 2,343 aerial images



V1.0

“After a hurricane, every hour without a damage map is an hour without a rescue plan.”

FINAL PROJECT · SPRING 2026

# Problem & Motivation

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After a hurricane, responders need to know water is - fast, and across hundreds of square miles of imagery.

- Manual inspection of drone imagery does not scale to disaster footprints.
- An automated **flooded vs. non-flooded** screen prioritizes ground crews.
- Same backbone can later extend to multi-class damage assessment.

1 hr

COST OF EVERY HOUR WITHOUT A MAP

2,343

AERIAL IMAGES ANALYZED

# Dataset: FloodNet-Supervised v1.0

High-resolution UAV imagery from Hurricane Harvey, paired with pixel-level semantic masks across **10 land-cover classes** .

| SPLIT      | IMAGES | MASKS | PAIRS |
|------------|--------|-------|-------|
| Train      | 1,445  | 1,445 | 1,445 |
| Validation | 450    | 450   | 450   |
| Test       | 448    | 448   | 448   |
| Total      | 2,343  | 2,343 | 2,343 |

Every image–mask pair successfully matched. No missing files.



Classes 1 (Building-flooded) and 3 (Road-flooded) are collapsed into a single binary **flood** indicator for image-level supervision.

# Task Definition

From a  $512 \times 512$  aerial RGB image, predict a single label: **flooded** or **non-flooded**.

- **Input.** Aerial RGB image, normalized with training-set channel statistics.
- **Label.** Image is *flooded* if the binary flood mask contains any positive pixel.
- **Loss.** Binary cross-entropy with sigmoid output.
- **Metrics.** Accuracy, precision, recall, F1, confusion matrix.

512<sup>2</sup>

INPUT RESOLUTION

10 → 2

CLASSES COLLAPSED

8

BATCH SIZE

42

RANDOM SEED

CLASS DERIVATION

flood = (mask == 1) OR (mask == 3)

1 = Building-flooded · 3 = Road-flooded

# Preprocessing Pipeline

01

Pair images with masks

Match .jpg to \_lab.png by filename stem; verify no missing missing pairs.

02

Resize to 512 × 512

Bilinear for images, nearest-neighbor for masks to preserve class IDs.

03

Normalize

Channel mean / std computed from training set only — no leakage.

04

Derive flood mask

Collapse classes 1 & 3 into a single binary mask, then smooth.

05

Image-level label

Label = 1 if any flood pixel exists; feed (image, label) to classifier.

CHANNEL STATISTICS (TRAIN)

|      | R      | G      | B      |
|------|--------|--------|--------|
| Mean | 0.4092 | 0.4471 | 0.3403 |
| Std  | 0.2013 | 0.1860 | 0.2025 |

WHY THIS MATTERS

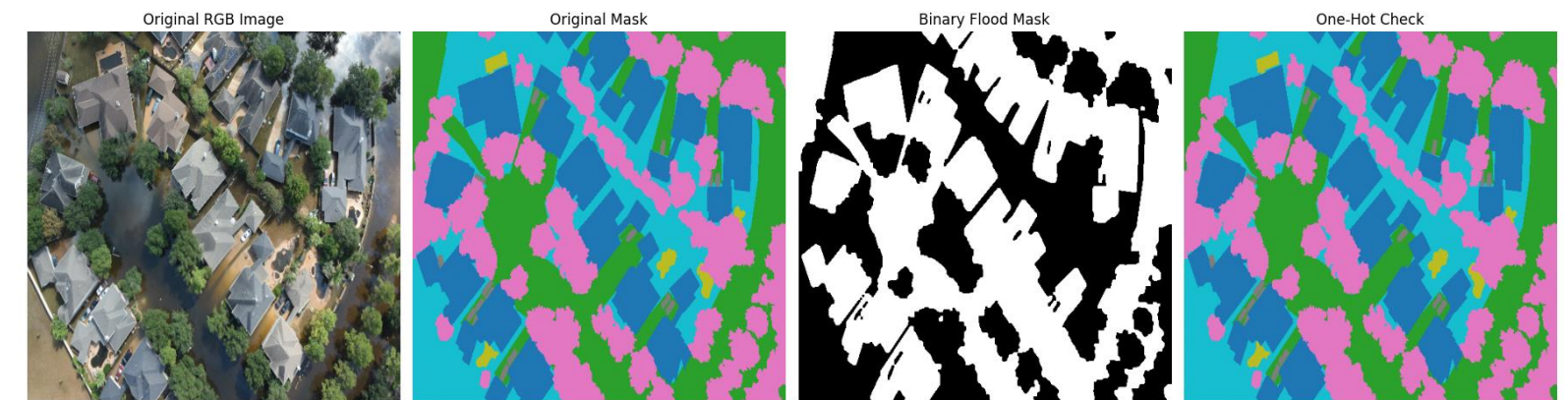
Computing statistics from the training split alone keeps validation and test sets genuinely held-out. Nearest-neighbor mask resizing prevents class IDs from being averaged into invalid intermediate values.



# Mask Smoothing

Raw flood masks are noisy. A short morphological pipeline cleans them before label derivation.

- 01 Bilateral filter.**  $d = 5$ ,  $\sigma_{\text{color}} = \sigma_{\text{space}} = 75$  — preserves edges, removes speckle.
- 02 Threshold.** Re-binarize back to  $\{0, 1\}$  after filtering.
- 03 Dilate  $\times 2$ .**  $3 \times 3$  kernel — close small gaps inside flooded regions.
- 04 Erode  $\times 1$ .**  $3 \times 3$  kernel — pull boundaries back, net dilation of one pixel.



OUTPUT OF THE SMOOTHING STEP ON A SAMPLE TILE

Binary flood mask (third panel) is the supervisory target the image-level label is derived from.

# Model Architecture

ImageNet-pretrained ResNet backbone, global average pooling, dropout, and a single sigmoid neuron for binary classification.



23.6 M

RESNET50 PARAMS

42.7 M

RESNET101 PARAMS

2,049

TRAINABLE (STAGE 1)

Adam

OPTIMIZER · BCE LOSS

# Two-Stage Training Strategy

STAGE 1 — FROZEN BACKBONE

## Train the head only

Backbone weights are frozen; only the dropout + dense layer learn. This adapts the ImageNet features to the flood task without destabilizing them.

|                  |       |
|------------------|-------|
| Epochs           | 20    |
| Learning rate    | 1e-3  |
| Trainable params | 2,049 |
| Optimizer        | Adam  |

STAGE 2 — PARTIAL UNFREEZE

## Fine-tune the last 30 layers

Unfreeze the final 30 backbone layers and continue at a much lower learning rate. Earlier layers stay fixed to preserve general visual features.

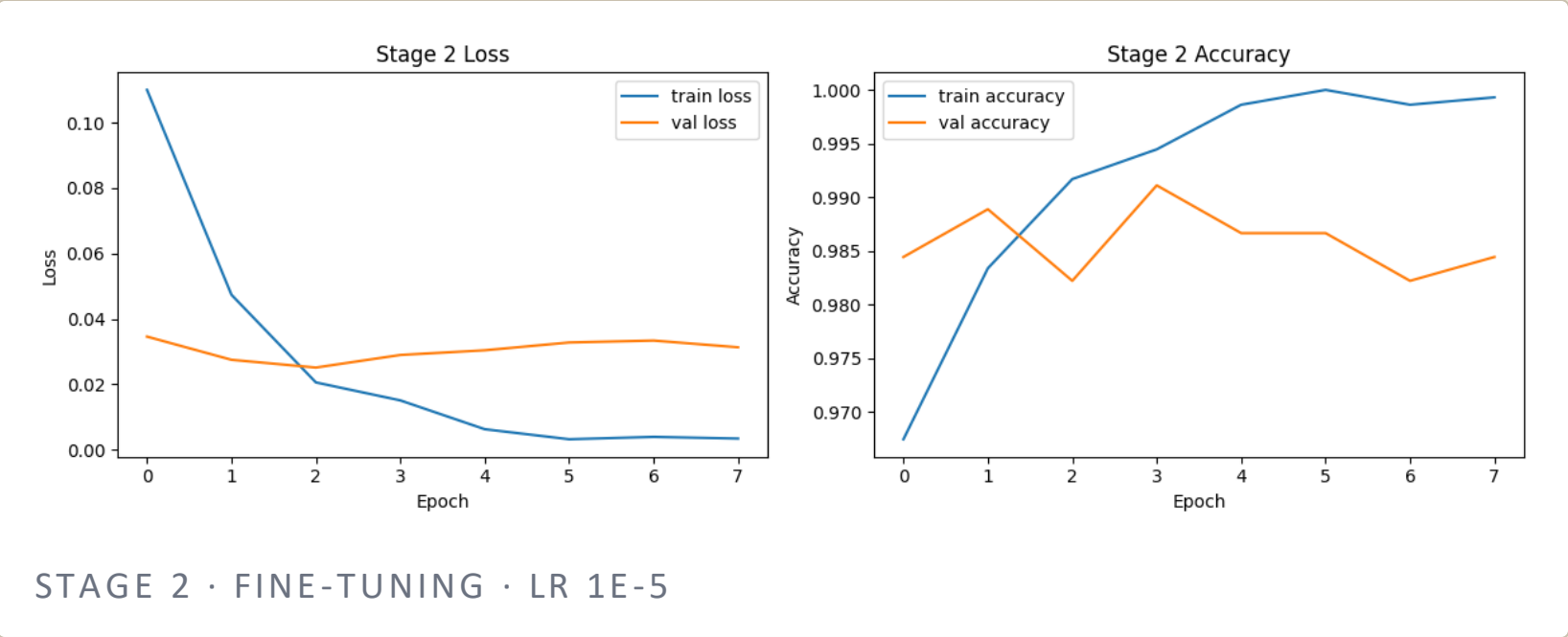
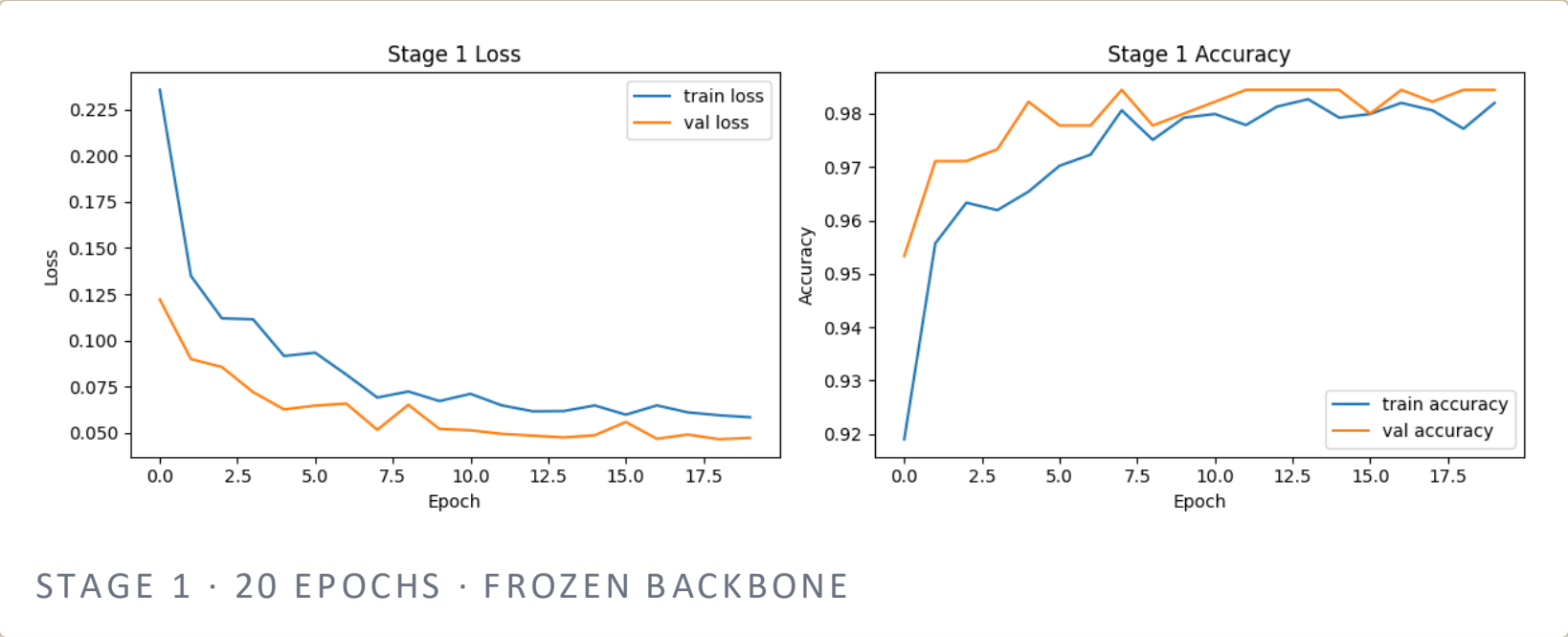
|                 |                   |
|-----------------|-------------------|
| Epochs          | 10                |
| Learning rate   | 1e-5              |
| Unfrozen layers | last 30           |
| Optimizer       | Adam (recompiled) |

CALLBACKS (BOTH STAGES)

**EarlyStopping** — monitor val\_loss, patience 5, restore best weights. · **ReduceLROnPlateau** — factor 0.5, patience 2.



# ResNet50 — Training Curves



Stage 1 climbs from 0.92 to ~0.98 validation accuracy and plateaus. Stage 2 fine-tuning pushes train accuracy toward 1.0 — early stopping restores the best weights at **val\_loss ≈ 0.025**.

0.025

BEST VAL LOSS

28

EFFECTIVE EPOCHS

# ResNet50 — Test Performance

99.55%

TEST ACCURACY

0.9808

F1 (FLOOD CLASS)

0.9808

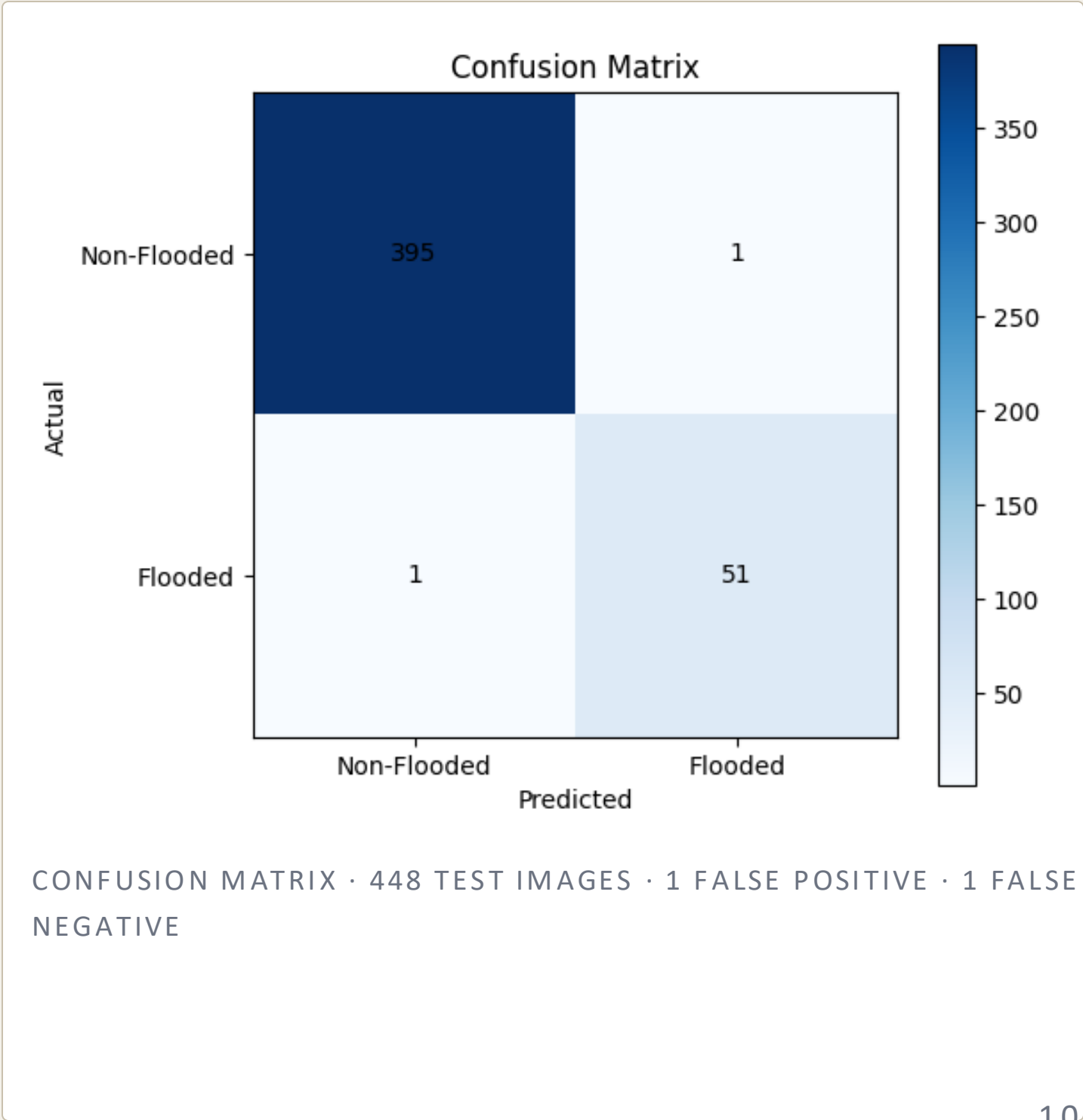
PRECISION · FLOODED

0.9808

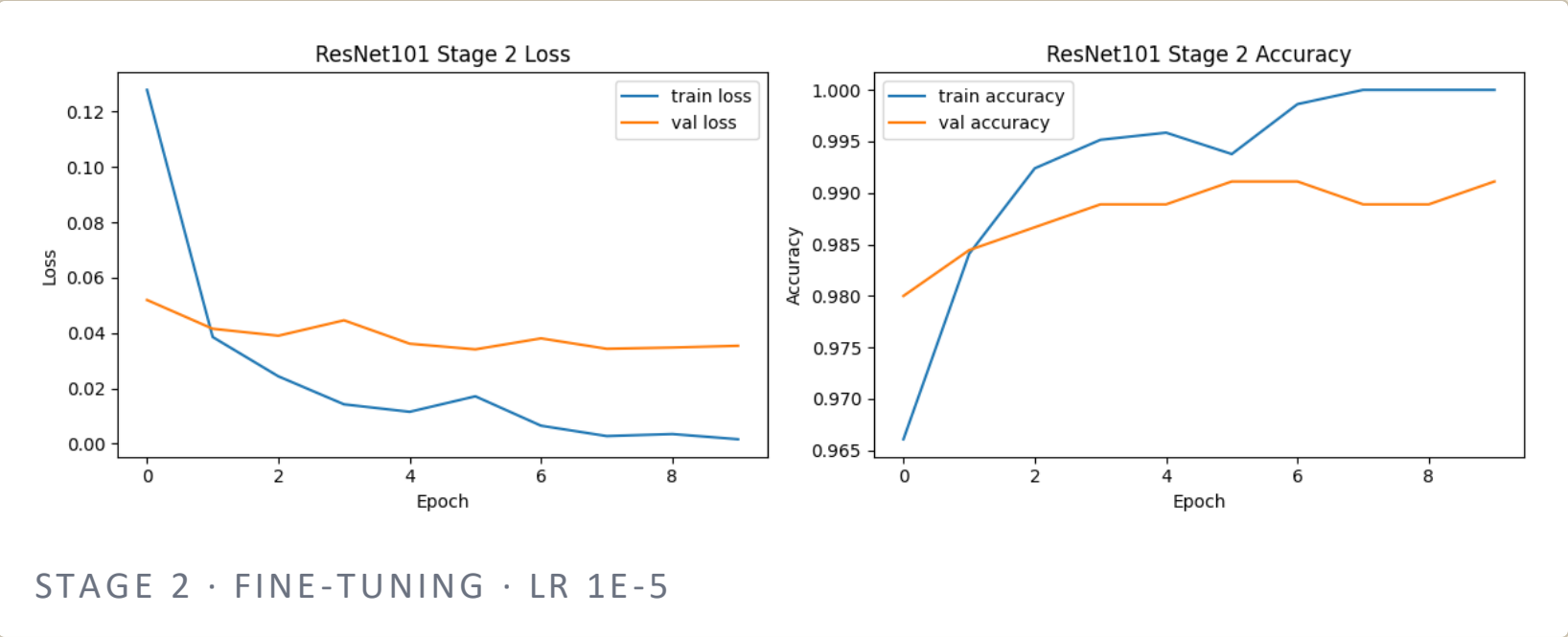
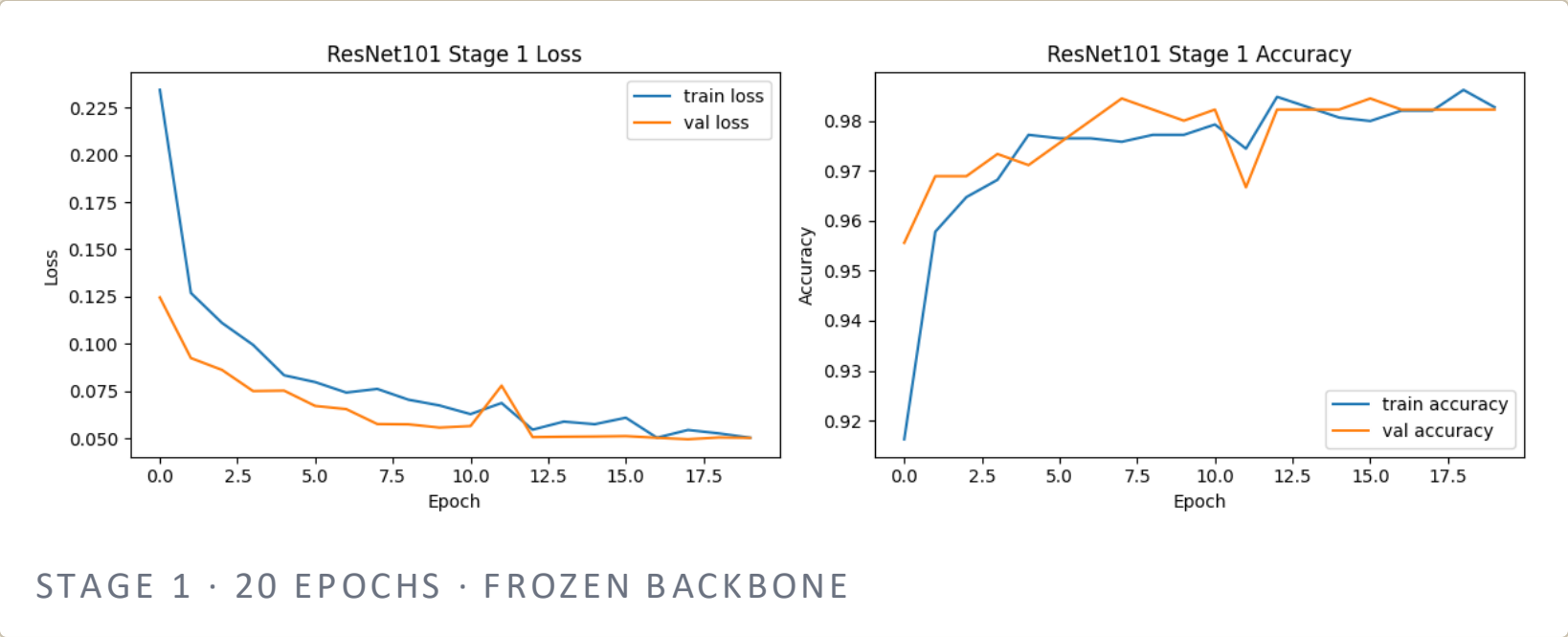
RECALL · FLOODED

| SPLIT      | LOSS   | ACCURACY | PRECISION | RECALL |
|------------|--------|----------|-----------|--------|
| Train      | 0.0043 | 0.9986   | 0.9939    | 0.9939 |
| Validation | 0.0251 | 0.9822   | 0.9259    | 0.9259 |
| Test       | 0.0194 | 0.9955   | 0.9808    | 0.9808 |

Test is the only unbiased estimate — val\_loss was monitored for early stopping.  
FLOOD DETECTION · FINAL PROJECT



# ResNet101 — Training Curves



ResNet101 reaches similar validation accuracy with a noticeably more stable Stage 2 — train and validation tracks diverge less than the ResNet50 run, consistent with deeper-network capacity used as a regularizer here.

0.991

BEST VAL ACCURACY

30

EFFECTIVE EPOCHS

# ResNet101 — Test Performance

99.55%

TEST ACCURACY

0.9804

F1 (FLOOD CLASS)

1.0000

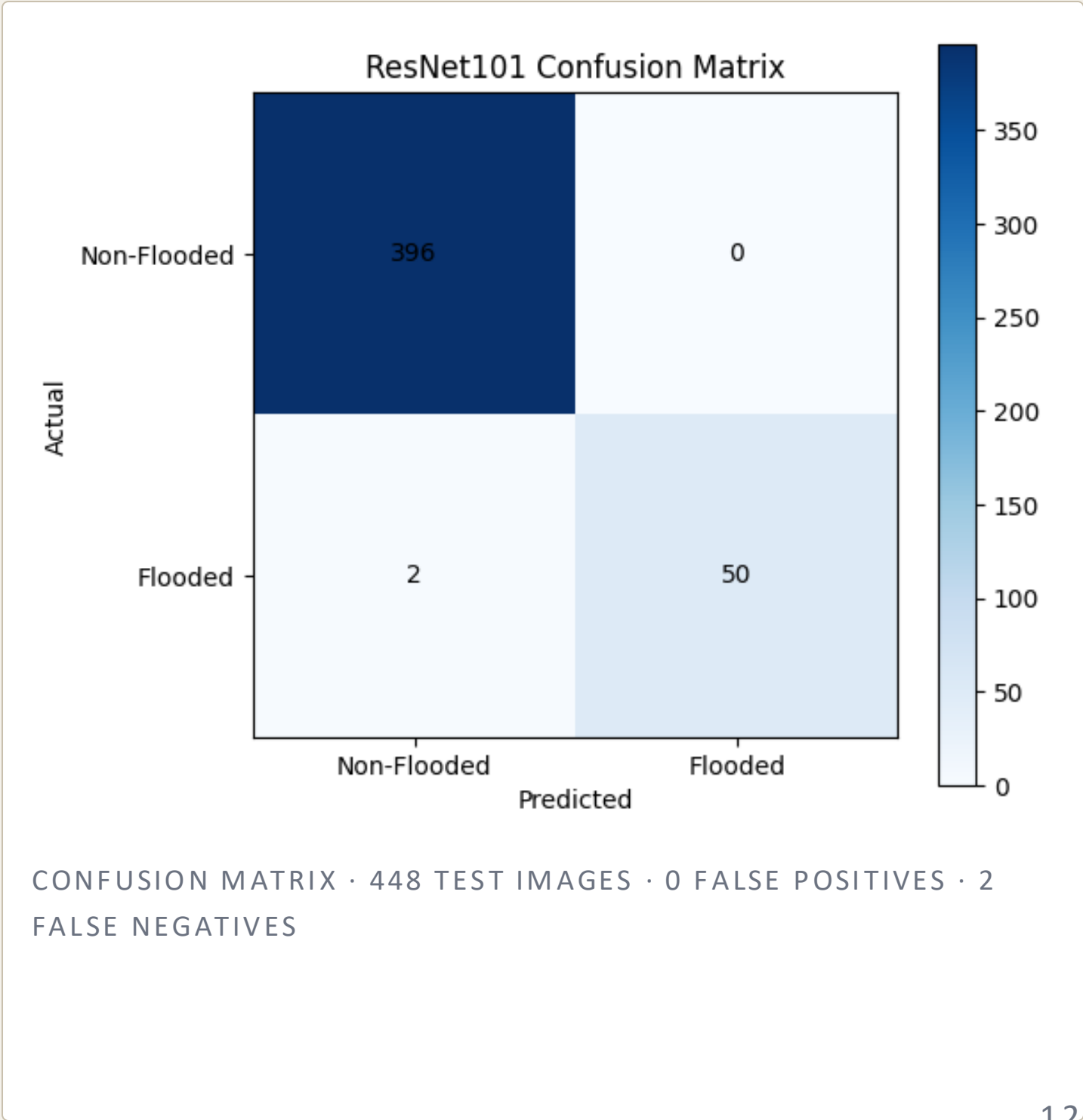
PRECISION · FLOODED

0.9615

RECALL · FLOODED

| SPLIT      | LOSS   | ACCURACY | PRECISION | RECALL |
|------------|--------|----------|-----------|--------|
| Train      | 0.0016 | 0.9993   | 0.9940    | 1.0000 |
| Validation | 0.0341 | 0.9911   | 0.9808    | 0.9444 |
| Test       | 0.0307 | 0.9955   | 1.0000    | 0.9615 |

ResNet101 trades two false negatives for zero false positives on test.



# Model Comparison

| MODEL     | TRAIN ACC | VAL ACC | TEST ACC | TEST F1 | TEST PRECISION | TEST RECALL |
|-----------|-----------|---------|----------|---------|----------------|-------------|
| ResNet50  | 0.9986    | 0.9822  | 0.9955   | 0.9808  | 0.9808         | 0.9808      |
| ResNet101 | 0.9993    | 0.9911  | 0.9955   | 0.9804  | 1.0000         | 0.9615      |

RESNET50

**Symmetric errors**

1 false positive, 1 false negative. Balanced precision / recall at 0.9808.

RESNET101

**Conservative — never cries flood**

0 false positives, 2 false negatives. Perfect precision; recall drops to 0.9615.

VERDICT

**Tie on accuracy**

Pick by use case: **ResNet50** for balanced screening, **ResNet101** when false alarms are costly.



## Key Findings

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- **Transfer learning works out of the box.** Both models cleared 99% test accuracy with under an hour of fine-tuning.
- **Two-stage training pays off.** Frozen-backbone warmup before partial unfreeze prevented the early-epoch instability typical of full fine-tuning.
- **Depth doesn't dominate here.** ResNet101 nearly doubles parameters but matches ResNet50's accuracy — the dataset is the ceiling.
- **The error trade-off is the real choice.** Symmetric (ResNet50) vs. precision-first (ResNet101) reflects the deployment policy more than model quality.

99.55%

TEST ACCURACY · BOTH MODELS

2 / 448

WORST-CASE ERROR COUNT ON TEST

# Limitations & Next Steps

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## LIMITATIONS

- Single hurricane (Harvey) — geographic and seasonal bias.
- Class imbalance: only 52 of 448 test images are flooded.
- Validation accuracy is optimistic — used for tuning, not estimation.
- Image-level labels discard rich pixel-level information.

## NEXT STEPS

- Promote to **semantic segmentation** using the same masks already prepared.
- Add data augmentation: flip, rotate, color jitter, weather synthesis.
- Cross-event evaluation on a different hurricane to test generalization.
- Deploy a sliding-window inference for full mission-area maps.

## References

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Li, Y., Martinis, S., & Wieland, M. (2019). Urban flood mapping with an active self-learning convolutional neural network based on TerraSAR-X intensity and interferometric coherence. *ISPRS Journal of Photogrammetry and Remote Sensing*, 152, 178-191.

Rahnemoonfar, M., Chowdhury, T., Sarkar, A., Varshney, D., Yari, M., & Murphy, R. R. (2021). Floodnet: A high resolution aerial imagery dataset for post flood scene understanding. *IEEE Access*, 9, 89644-89654.



*Thank you.*

Questions, critiques, and ideas for extending this beyond a single hurricane are all welcome.